the amplitude of the update step in the problem space is not constrained (so we could move even between opposite corners of the domain), and take a look at figure 17.6.

Initially, when the temperature of the system is high (figure 17.6.A), and by construction the probability distribution should allow almost any update (no matter how bad it seems), the algorithm can move all over the landscape and even get out of local minima: even if this means that it might happen to abandon an optimal position for one of the worst negative peaks. Actually, at this stage, it is even easy for the algorithm to “walk away” from good solutions.